

Exercise 11

$$\underline{1.1} \quad \text{MSE}(m, b) = \frac{1}{N} \sum_{i=1}^N (mx_i + b - y_i)^2$$

$$\frac{\partial \text{MSE}}{\partial m} = \frac{2}{N} \sum_{i=1}^N x_i (b + mx_i - y_i)$$

$$\frac{\partial \text{MSE}}{\partial b} = \frac{2}{N} \sum_{i=1}^N (b + mx_i - y_i)$$

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$$\sum_{i=1}^N y_i - m \sum_{i=1}^N x_i + bN = 0$$

$$b = \frac{\sum_{i=1}^N y_i - m \sum_{i=1}^N x_i}{N}$$

Then I put this into the first equation and solve for m :

$$\sum_{i=1}^N x_i y_i - m \sum_{i=1}^N x_i^2 - \left(\frac{\sum_{i=1}^N y_i - m \sum_{i=1}^N x_i}{N} \right) \sum_{i=1}^N x_i = 0$$

$$\Rightarrow m = \frac{N(\sum x_i y_i) - (\sum x_i)(\sum y_i)}{N(\sum x_i^2) - (\sum x_i)^2}$$

$$b = \frac{(\sum y_i) - m(\sum x_i)}{n}$$

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